

Homework 5

Andrew Hwang

Math 38

Gayee Park

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Question 1. Read the proof of Theorem 3.1.16 in the textbook (or from another source). In your own words summarize the proof idea.

Theorem (König–Egerváry, 1931). If G is a bipartite graph, then the maximum size of a matching in G equals the minimum size of a vertex cover of G .

Proof. König–Egerváry’s Theorem relies on the relationship between matchings and augmenting paths. Let M be a maximum matching in a bipartite graph G with bipartition X, Y . To show $|M| = \beta(G)$, we construct a vertex cover U of size $|M|$. The strategy involves exploring G starting from M -unsaturated vertices in X using M -alternating paths. Let $S \subseteq X$ and $T \subseteq Y$ be the sets of vertices reachable from unsaturated vertices in X via such paths. By Berge’s Theorem, no alternating path can end in an unsaturated vertex in Y , or it would be an augmenting path. We define the vertex cover as $U = (X \setminus S) \cup T$.

- **Size:** Each edge in M contributes exactly one vertex to U (either an endpoint in T or an endpoint in $X \setminus S$), and no unsaturated vertices are in U , so $|U| = |M|$.
- **Covering:** Every edge must be covered: edges within the searched forest are covered by T , and edges connected to the “unreached” part of X are covered by $X \setminus S$.

Since $\alpha'(G) \leq \beta(G)$ always holds, finding a cover where $|U| = |M|$ proves the equality. $\square$

Question 2. Prove that every maximal matching in a graph G has at least $\alpha'(G)/2$ edges.

Proof. Let M be a maximal matching and M^* be a maximum matching in G , such that $|M^*| = \alpha'(G)$.

By the definition of maximal matchings, no more edges can be added to M . This implies that every edge $e \in M^*$ must be incident to at least one edge in M . If there were an edge $e = \{u, v\} \in M^*$ such that neither u nor v was covered by M , then $M \cup \{e\}$ would be a larger matching, contradicting the maximality of M .

Let $V(M)$ be the set of vertices covered by M . Since each edge in M corresponds to two distinct endpoints, $|V(M)| = 2|M|$. As established, every edge in M^* must have at least one endpoint in $V(M)$. However, each vertex in $V(M)$ can be incident to at most **one** edge in M^* (because M^* is a matching). Therefore,

$$|M^*| \leq |V(M)|.$$

Substituting $|V(M)| = 2|M|$ gives

$$|M^*| \leq 2|M| \implies \boxed{|M| \geq \frac{|M^*|}{2} = \frac{\alpha'(G)}{2}}.$$

□

Question 3. Prove that $\alpha(G) \geq \frac{n(G)}{\Delta(G)+1}$ for all graphs G , where $n(G) = |V(G)|$, $\alpha(G)$ is the independence number, and $\Delta(G)$ is the maximum degree of G .

Proof. We can prove this using a **greedy algorithm** construction. Let V be the vertex set of G .

Initialize:

Construct an independent set I .

Iteration:

1. Pick an arbitrary vertex $v_1 \in V$. Add v_1 to I .
2. Remove v_1 and all its neighbors $N(v_1)$ from the graph.
3. Repeat the process by picking a vertex v_2 from the remaining graph, adding it to I , and removing v_2 and $N(v_2)$.
4. Continue until no vertices remain.

In each step i , we add exactly 1 vertex to I . The number of vertices removed from the graph in step i is $1 + d(v_i)$, which is at most $1 + \Delta(G)$. Let k be the total number of vertices in the resulting independent set I . The total number of vertices removed across all k steps must equal $n(G)$. Therefore,

$$n(G) \leq \sum_{i=1}^k (1 + \Delta(G)) = k(1 + \Delta(G)).$$

Rearranging for k , we get

$$k \geq \frac{n(G)}{\Delta(G) + 1}.$$

Since $\alpha(G)$ is the size of the *maximum* independent set, and we have found one independent set of size k ,

$$\alpha(G) \geq k \geq \frac{n(G)}{\Delta(G) + 1}.$$

□